RESEARCH REPORT / VISUAL MODEL MAINTENANCE

EditGuard

A reversible workbench for visual model corrections

Published model editing, bounded fine-tuning, and retrieval compared using generated answers.

Measured experiment	Completed scope	Inference rule
20 edit lineages	20 edits x 5 methods x 5 cases = 500 records	Greedy generation, 24-token cap

The measured tradeoff. Adapted FT matches all 60/60 edited-question targets, but answers 0/40 unrelated questions correctly and loses all 5 previously correct locality answers. MEND retains 5/5 of those answers, with lower target success. Five retained facts are too few to establish broad safety.

What was built

A local workbench wraps the published VLKEB/MEND research code and BLIP-2 model. It can propose an actual parameter update, save an inactive candidate, show generated before/after answers and an unrelated probe, require explicit operator approval, and restore original weights exactly.

What was measured

The saved final report records **20/20 completed edit units**, 0 unexecuted units and a completed worker. It compares base, prompt only, updated visual RAG, adapted MEND and adapted last-layer FT. All assigned questions remain in denominators.

How to read this report

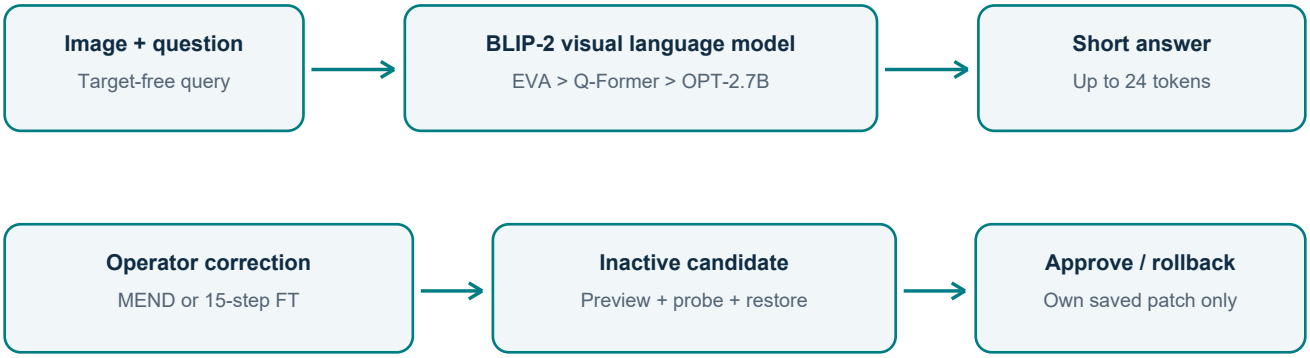
Pages 2-4 explain the model and the fixed experiment. Pages 5-7 show results, side effects and timing. Pages 8-11 explain failure cases, the application, reproduction and limits. Published paper numbers are not inserted into our result tables.

Prepared 24 October 2026, 12:43 UTC. Measurements are read from a saved JSON report; this document runs no model or training.

EDITGUARD / 02

From an image to a short answer

The base model combines a visual encoder and a language model. Editing changes selected internal weights; retrieval changes the information supplied to an unchanged model.



Component	Beginner's explanation	Who supplied it
EVA image encoder	Turns the photograph into visual features.	Published pretrained weights
Q-Former	Condenses visual features into vectors usable by the language model.	Published BLIP-2 components
OPT-2.7B	Produces the answer one token at a time.	Published pretrained weights
MEND transform	Uses an edit gradient to propose a targeted weight update.	Published pretrained editor
Our workbench + evaluator	Adapters, experiment controls, saved evidence, approval and exact restoration.	This project

Attribution boundary. We did not pretrain the foundation model or meta-train MEND. FT is real per-correction optimization; using a pretrained editor is a different kind of work.

The standard query contains the image and question. No expected answer is supplied to the ordinary base/MEND/FT generation call. Prompt and RAG baselines intentionally receive the current correction as a demonstration, which is their declared behavior.

Terms used here: weights are the model's stored numerical parameters; a layer is one processing stage. A patch saves changes to selected weights. A token is a piece of text, so 24 tokens need not mean 24 words.

Five ways to handle one correction

FT means fine-tuning: optimize selected weights for the correction. RAG means retrieval-augmented generation: supply a relevant saved correction in the input. MEND is a published editor that transforms a gradient into a weight update.

Method	What changes?	What the query receives
Base	Nothing	Image + question
Prompt only	Input prefix; weights fixed	Current correction as a QA example, then the actual query
Updated visual RAG	One-correction memory; weights fixed	QA example when image/question scope score accepts; otherwise base prompt
MEND (adapted)	Six OPT MLP matrices in layers 29-31	Actual query after one learned-transform parameter update
FT (adapted)	Two OPT layer-31 matrices	Actual query after 15 AdamW steps

Adaptations are part of the comparison

Question format: every query uses *Question: ... Answer:*. **Context-aligned labels:** target labels are taken from the actual joined question/target tokenization. The development diagnosis found six standalone target tokens but four after a separating space; assuming the standalone count can supervise the wrong positions.

FT: 15 AdamW steps at learning rate 0.0002 update layer 31 fc1/fc2 only. FP32 master optimizer copies retain the released base precision. A supervised newline teaches termination. MEND retains its context-aligned target-only objective, so objectives are not identical.

RAG: scope combines MiniLM question cosine and BLIP visual-prefix cosine equally. A threshold selected on TRAIN rows 2-9 gates the current correction. The encoders are frozen; the score is not a calibrated probability. Text-only locality is outside the visual memory scope and uses the base model.

These are documented research adaptations. The experiment does not reproduce every paper setting or isolate editor architecture under identical objectives.

EDITGUARD / 04

A fixed, finite evaluation

Twenty edit lineages are selected before model inference, with five linked questions per lineage. Each method receives the same assigned cases.

Case	What it tests	Assigned per method
Direct	The requested target on the correction image/question	20
Paraphrase	Transfer to the authors' alternate wording	20
Other photo	Transfer to another photograph of the source entity	20
Text locality	An unrelated question with no image	20
Visual locality	A similar but unrelated image/question	20

The source is the authors' benchmark, not a new independent dataset. Source image-directory IDs found in any training edit, alternate-image or visual-locality slot are excluded. Stable lineage hashes determine the first 20 distinct source entities. This is a small nonrandom slice, not a random population sample.

Quantity	Meaning
Strict accuracy	Entire generated answer matches the target after fixed normalization. A target merely mentioned in repetitive prose is insufficient.
Abstention / error	Empty answers, declared nonanswers, runtime errors and missing cases remain failures in assigned denominators.
Locality correctness	An unrelated query still receives its factual target answer.
Preservation	The answer stays verbatim unchanged. Conditional retention uses only originally correct base answers.
Paired interval	95% bootstrap interval over complete editing units, using 10,000 resamples; linked questions stay together.

The source, selection, threshold, code, scorer and settings were sealed before final inference. No post-test answer aliases or scoring changes are introduced by this PDF.

Measured generated-answer results

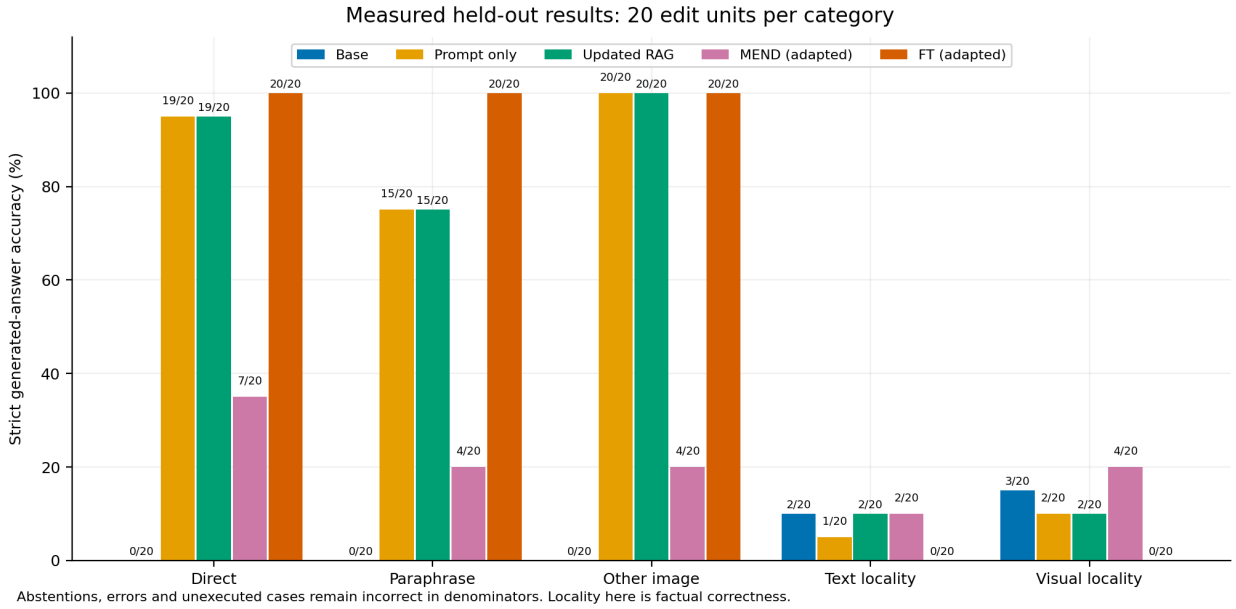


Figure 1. Saved final measurements, 20 edits per cell. Locality columns measure factual correctness; empty/error cases remain in denominators.

Method	Direct	Para.	Other photo	Text local	Visual local	Abstain	Error
Base	0/20	0/20	0/20	2/20	3/20	4	0
Prompt only	19/20	15/20	20/20	1/20	2/20	0	0
Updated RAG	19/20	15/20	20/20	2/20	2/20	1	0
MEND (adapted)	7/20	4/20	4/20	2/20	4/20	2	0
FT (adapted)	20/20	20/20	20/20	0/20	0/20	0	0

Read both sides of the tradeoff. FT matches 60/60 edited targets and 0/40 unrelated targets. RAG and prompt only each match 54/60 edited targets. MEND matches 15/60 while retaining the 5 previously correct unrelated answers.

Base's 0/60 is agreement with the benchmark's counterfactual replacement targets, not its general factual accuracy. Each cell has 20 cases: one match changes accuracy by five percentage points. Abstain and Error columns aggregate over the 100 assigned outcomes per method; empty answers remain failures.

Scope, side effects and uncertainty

Method	Text facts retained	Visual facts retained	All facts retained	Verbatim preserved
Base	2/2	3/3	5/5	5/5
Prompt only	0/2	2/3	2/5	1/5
Updated RAG	2/2	2/3	4/5	3/5
MEND (adapted)	2/2	3/3	5/5	5/5
FT (adapted)	0/2	0/3	0/5	0/5

Small denominator. The base answered only 5/40 unrelated questions correctly: two text and three visual. Retention above concerns only those 5 correct answers. MEND's 5/5 does not establish reliable locality on a broader population. Unchanged wrong answers are not factual success.

Factual retention uses normalized whole-answer scoring. Verbatim preservation requires identical output text, so capitalization alone can make the two counts differ. Unconditional preservation, including originally wrong answers, is separately recorded in the JSON.

Updated RAG: routing is separate from answer quality

Accepted in scope	Missed in scope	Applied outside scope	Correctly rejected
60/60	0/60	13/40	27/40

Routing precision: 82.2%; applicable recall: 100.0%; false application rate: 32.5%. Routing errors/unexecuted: 0. A correct route does not guarantee a correct generated answer.

Selected paired differences

Comparison	Difference	95% paired interval
MEND (adapted) minus Base (Direct)	+35.0 pp	[+15.0, +55.0] pp
MEND (adapted) minus FT (adapted) (Visual locality)	+20.0 pp	[+5.0, +40.0] pp
Updated RAG minus MEND (adapted) (Direct)	+60.0 pp	[+35.0, +85.0] pp
Updated RAG minus FT (adapted) (Paraphrase)	-25.0 pp	[-45.0, -10.0] pp

pp means percentage points. Saved 95% intervals use 10,000 paired edit-unit bootstrap resamples (seed 42). These descriptive examples are not adjusted for multiple comparisons. All 50 comparisons remain in the JSON and paired-interval figure. Resampling this slice cannot remove selection bias or establish performance on unseen entities.

Time and memory are separate outcomes

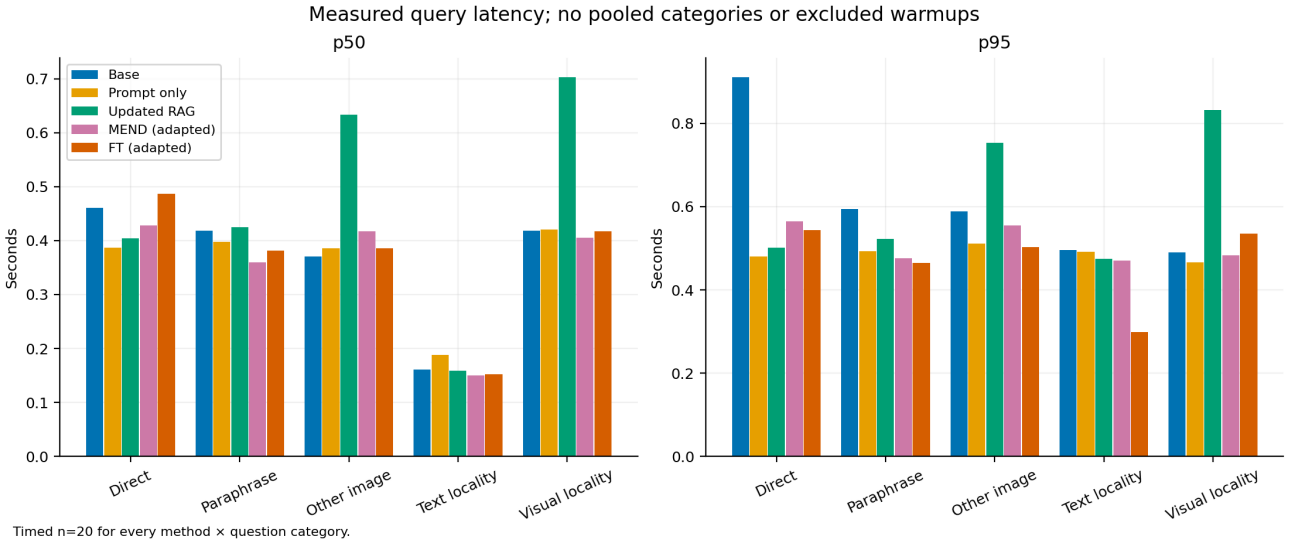


Figure 2. Query latency from saved measurements. Categories are kept separate; sample counts and the exact timing rule are recorded below.

Operation	Timed n	p50 seconds	p95 seconds
MEND (adapted) edit	20	3.553	3.645
FT (adapted) edit	20	10.242	10.427
RAG reference index	20	0.265	0.298

CUDA synchronize before/after each call; zero excluded warmups; includes errors with recorded durations; loading/editing separate from query latency; p95 linear interpolation; no cross-category pooled latency

Per-category query sample counts range from 20 to 20. p50 is the median; p95 is an interpolated tail summary, not a worst-case guarantee. Editing and indexing are not hidden inside ordinary query latency.

Worker resource	Saved measurement
Worker elapsed	815.078 seconds
Sampled process RSS peak	7.867 GiB
CUDA allocated / reserved peaks	11.152 / 11.711 GiB
Recorded device	Tesla T4

RSS sampling can miss brief peaks, and CUDA allocator values omit some driver allocations. These measurements do not establish fit on the laptop's 6 GB GPU or predict another host's latency. One GiB is 1,073,741,824 bytes.

Why direct success needs review

The following are saved final-experiment cases, inspected retrospectively to explain the aggregate results. They are not new confirmatory tests.

Saved final case	Observed behavior	Why it matters
Football club; replacement target Wellington Phoenix FC	FT matches all three edited questions. On an unrelated logo, base/MEND answer SpVgg Unterhaching; prompt/RAG answer Wellington Phoenix FC and FT repeats Wellington.	Target success can coexist with loss of a previously correct answer.
Movie; replacement target Hudson Hawk	RAG accepts the paraphrase route but generates The Mask. FT changes the unrelated Jurassic Park answer to Hudson Hawk.	Correct routing need not produce the requested answer; editing may spread beyond its scope.
Unrelated text question; target The Simpsons	Base/RAG/MEND answer The Simpsons. Prompt only gives The Big Bang Theory; FT gives Mark Goodson.	An image correction can affect a text-only query.

Case IDs start 0051675ae507, 022ff831f6c4 and 0392b4baabf3 in the saved per-edit JSON directory. Replacement targets are controlled benchmark labels, not independently verified corrections.

Development checks establish mechanics

Earlier TRAIN-fixture MEND and FT runs each passed 8/8 mechanical checks, including fresh-process patch reload and exact restoration. FT generated the direct Albany, New York target, but its paraphrase was only Albany and its unrelated answer repeated Albany. The unrelated base answer was already empty and wrong; this development case demonstrates changed behavior, not loss of a correct fact.

Why a perfect token score was not enough

The loaded MEND patch scored 100% on contextual teacher-forced target tokens for the raw direct question, yet freely generated repeated city text. Teacher forcing places the full target in the model input; free generation must choose its own next tokens and stop. They answer different questions and cannot share an accuracy claim.

A failure receipt is still evidence

The first raw/release MEND attempt changed weights and restored them in-process, but its reload worker was killed with return code -9. The complete persistence gate failed. Memory pressure is plausible, but the return code alone does not prove an out-of-memory kill. Missing phase measurements are not replaced with estimates.

Development fixtures came from the authors' TRAIN annotations and may have been seen during editor training. The final cases above are separate. Neither the mechanical checks nor a selected anecdote estimates general reliability.

A reviewable, reversible application

Stage	What the operator sees	Weight state
Generate	Actual current answer, selected source and image	Original or explicitly active state
Preview	Before/candidate answers, target match and one fixed unrelated probe	Candidate tested, then exact base restoration
Review	Saved question, target, outputs, provenance and hashes	Inactive candidate
Approve + activate	Explicit operator decision, including any visible warnings	Own saved patch checked and loaded
Rollback	Restoration result and updated evidence history	Saved original tensors copied back

Only one patch may be active. Another preview or activation is rejected until rollback; the UI disables those actions. Clients cannot supply arbitrary patch paths or checkpoint uploads. POST operations require a per-process token. These are local academic controls, not production multi-user authentication.

The server serializes model operations and binds only to 127.0.0.1. A fresh successful model load starts with original weights and marks formerly active history inactive_after_restart. Historical patches are not silently reapplied. Keep history and exported patches on persistent storage.

Evidence boundary

Eleven application contract tests use a tiny fake backend to check state transitions and failure recovery. They do not establish real-model quality. GitHub CI also built the pinned Linux container, checked offline imports and the CLI, and verified runtime-user history writes. That is CPU packaging evidence, not a full GPU server deployment.

Separate deployment check. A real-GPU browser/API lifecycle rehearsal is required in addition to experiment receipts. No publicly hosted application or GPU Docker runtime is claimed by this PDF.

The evaluation dashboard reads a configured local report unchanged. If it is absent, numerical cards are absent. Downloadable evidence makes the displayed claims inspectable; paper results cannot fill a missing local measurement.

Reproduce the evidence, not a promise

Step	Concrete artifact or command
Pinned environment	Python 3.11.9; torch 2.0.1+cu117; exact dependency closure in requirements-vlkeb.txt
Verified assets	25 declared base/editor/fixture inputs; 13,218,483,144 bytes, plus separate pinned MiniLM files
Setup	python scripts/bootstrap_vlkeb_colab.py --fetch-upstream --install
Prepare	Explicit asset download, verification and deterministic selection; fixed threshold-validation receipt
Seal then run	python -m ke_repro.evaluate --seal ; python -m ke_repro.evaluate --run
Finite cap	1,800-second worker cap per attempt; same frozen seal for an interrupted-run resume
WorkBench	artifacts/vlkeb/environment/bin/python -m ke_repro.workbench --port 8003

The dependency environment is isolated from notebook Python and the Windows tools. Model inference uses verified local files in offline mode. The one-episode planning estimate was approximately 15.5 minutes; it was not a runtime guarantee. No editor meta-training is performed. A complete final attempt requests at most 20 MEND updates and 300 FT optimizer steps.

Container and viewer

On a compatible native Linux NVIDIA host, Docker uses --gpus all and --network host because the application stays on loopback. Mount assets read-only and history writable with the host user ID. GPU container execution remains a separate runtime check. Colab's official notebook port viewer depends on its active runtime and access; it is not public hosting.

What to preserve

Keep the seal, selection checksum, attempt logs, worker report, per-edit rows, evaluation JSON, figure CSV/JSON and source hashes. Model weights and approximately 600 MiB patches remain outside Git. A patch hash identifies a patch but cannot reconstruct missing tensor data.

Rights are component-specific: VLKEB/editor, OPT, LAVIS, MiniLM, annotations and photographs have separate terms. OPT's noncommercial research restriction is not removed by another component's permissive code license.

Limits, references and provenance

Scope: 20 nonrandom public benchmark edits and one-edit retrieval memory do not establish arbitrary private-item recognition, open dialogue, lifelong composition or large-memory scalability. Source-directory separation does not rule out foundation-model pretraining or target/locality overlap.

Labels and scoring: authors' labels receive automated structural/checksum review, not independent human semantic review. Whole-answer matching can reject reasonable partial answers, and some locality references are long relative to the 24-token cap. This is not unrestricted semantic QA accuracy. Counterfactual targets are benchmark instructions, not newly verified facts.

Comparisons: contextual labels, prompt formatting, loading precision and FT termination supervision are disclosed adaptations. The prompt-only path lacks a second reference image; MEND and FT objectives differ. Published teacher-forced numbers are not directly comparable with generated-answer scores.

Operations: one fixed preview probe and exact rollback support controlled investigation. They do not guarantee harmless changes, multi-user security or production readiness. GPU memory and timing are host- and workload-specific.

Primary references

1. VLKEB paper: benchmark and published results
2. MEND: learned model editing
3. Pinned VLKEB source
4. Pinned benchmark dataset
5. BLIP-2 architecture
6. Pinned MiniLM sentence encoder
7. Project source and evidence

Auditable inputs

Final evaluation SHA256: db9e89cdfad68841de99a5e185aacd661fdf2afee07409a9dd774b400e2d433d

Sealed experiment SHA256: 73c303fd260977f624bf39de1d4f58fb8d9f9e2a8ae70f9aeca2fab72292a32f

Final inference source checkout: 460a91c6afbb8cb50857c275e4a913f06e135df7

Figure data SHA256: a8c168b1b285aa72de8e451a4e712f0a5483d36234ace8afb2d6970299bc14ee

Presentation builder checkout: ff032fc9842023dd4495781c73ea50072d67e93c

Presentation builder SHA256: 0eeffcd2dc6203ed616c3b1dbef906bccfa6519f7e5bee1303d9c60fe408e909

Project source documents: VLKEB_RESULTS.md, VLKEB_MODEL_CARD.md, VLKEB_DATASET_CARD.md, VLKEB_ARCHITECTURE.md, VLKEB_DEMO.md, VLKEB_WORKBENCH.md, VLKEB_EVALUATION_PROTOCOL.md, VLKEB_UPSTREAM_RESULTS.md.

The PDF reads the completed report and source-bound figures. It does not recompute model outputs, revise the frozen protocol or introduce missing measurements. Complete per-category latency, paired intervals and raw outcomes remain in the machine-readable evidence.